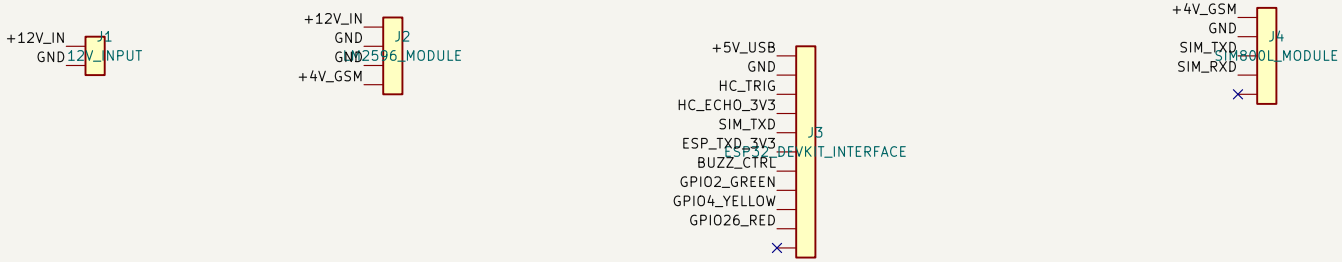
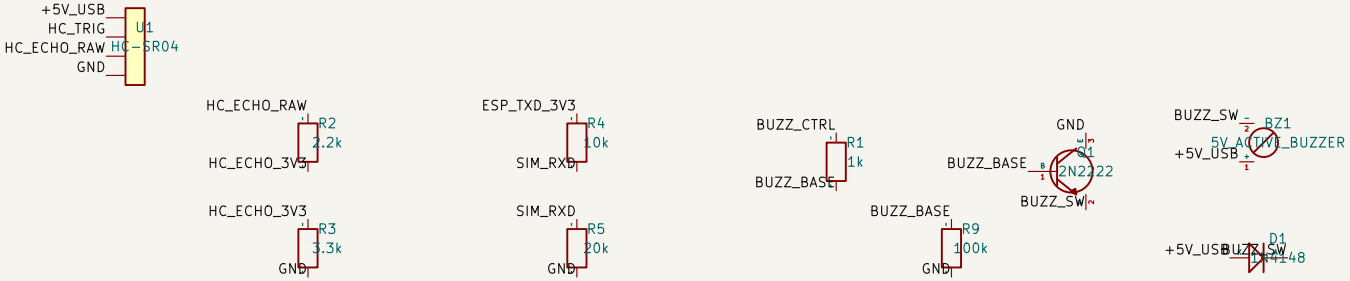
SMART SEWAGE MONITORING SYSTEM — MODULE CARRIER

ESP32 USB/VIN supplies the HC-SR04 and buzzer. The 12 V input feeds an external LM2596 set to 4.0 V for SIM800L.

J3 pin order: 5V, GND, GPIO5, GPIO18, GPIO16, GPIO17, GPIO25, GPIO2, GPIO4, GPIO26, NC.

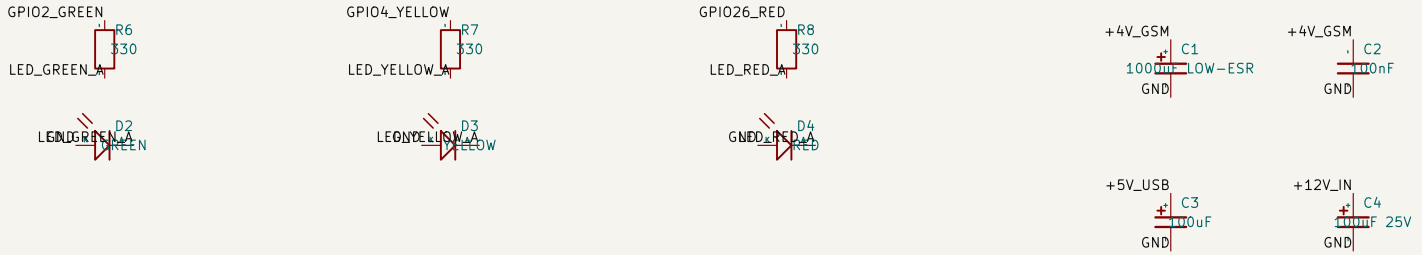
POWER / MODULE INTERFACES





LEVEL SHIFTING AND BUZZER DRIVER

STATUS INDICATORS AND DECOUPLING



Fit C1 close to J4. Verify 4.0 V at J4 pin 1 before installing the SIM800L module.

All module grounds are common. J3 is a labeled cable header; verify the pinout of the specific ESP32 DevKit used.

Set LM2596 output to 4.0 V before connecting SIM800L

ESP32 DevKit + HC-SR04 + SIM800L + external LM2596

SSEMAS

Sheet: /

File: schematic.kicad_sch

Title: Smart Sewage Monitoring System — Module Carrier

Size: A4	Date: 2026-10-29	**Rev: 2.0**
KiCad E.D.A. 10.0.7		Id: 1/1